Akoizo Original Practice - Form 4

Division B | Roller Coaster | 50-minute written practice

Original Akoizo curriculum practice. This is not a historical tournament paper. Written build, CAD, lab and communication exercises supplement real practical preparation.

Name: _____ Date: _____

1. (2 points) An elevator has IMA 0.5. If the counterweight drops 6 cm, how far can the ball rise?

A. 6 cm

B. 0.5 cm

C. 12 cm

D. 3 cm

Answer: _____

2. (2 points) What happens if a timing section's slope is too gentle?

Unit: Energy, friction, and the rolling ball.
Worked-case data: A solid ball drops 0.40 m from rest without losses. Find its speed.
Separate investigation observations:
Predicted speed: 2.4 m/s at the bottom of a 0.40 m drop.
Measured speed: 2.0 m/s.
Track: Three joints along the drop.

A. Nothing changes

B. The loop height increases

C. The ball may stop, ending timing

D. The ball speeds up

Answer: _____

3. (2 points) Which action is NOT allowed during a scorable run?

Unit: Testing and competition day.
Worked-case data: Target 52 s. Practice run: 50.6 s. The calibration table says each screw turn adds 0.4 s. What should the team do?
Separate investigation observations:
Test week 1: Loop failed 3 of 10 runs.
Cause: Ball hit a raised joint just before the loop.
After sanding joint: Loop worked 10 of 10.

A. Touching the device

B. Removing the pencil on “Go”

C. Watching the ball

D. Standing near the device

Answer: _____

4. (2 points) With a 45 s Target Time, which run time earns the most Time Score?

A. 45.5 s

B. 44.9 s

C. 46.2 s

D. 43.0 s

Answer: _____

5. (4 points) Explain why the ball must have a minimum speed at the top.

Unit: Loops: physics and scoring.
Worked-case data: A loop has a 9.0 cm path radius. What is the ideal minimum release height above the loop bottom, and a practical target?
Separate investigation observations:
Release 24 cm: Ball fell from top of loop in 6 of 10 trials.
Release 28 cm: Ball completed loop 10 of 10.
Loop path radius: 9 cm.

Answer: _____
_____
_____

6. (4 points) What happens to scoring if the ball stops before the Finish Line?

Answer: _____
_____
_____

7. (4 points) What should a safety stop do?

Unit: Size and construction.
Worked-case data: Option 1: 55 x 25 x 75 cm with a 22 cm loop. Option 2: 45 x 20 x 65 cm with a 15 cm loop. Compare Size + Loop points.
Separate investigation observations:
Home runs: Consistent 44.7 s.
After car ride: Runs 42.1 s; base slightly twisted.
After leveling: Runs 44.6 s.

Answer: _____
_____
_____

8. (4 points) Why must elevators be tested many times?

Unit: Elevators: levers, wheels, and pulleys.
Worked-case data: A second-class lever has IMA 1.6. It drives a movable pulley inverted to give IMA 0.5. Is the combination legal, and how far does the ball rise if the effort moves 8 cm?
Separate investigation observations:
Elevator test: Counterweight drops 5 cm; ball rises 9 cm.
Requirement: Ball must rise at least 10 cm.
Team claim: Elevator earns 50 points.

Answer: _____
_____
_____

9. (4 points) Why might a steel ball be more consistent than a plastic one?

Unit: Energy, friction, and the rolling ball.
Worked-case data: A solid ball drops 0.40 m from rest without losses. Find its speed.
Separate investigation observations:
Predicted speed: 2.4 m/s at the bottom of a 0.40 m drop.
Measured speed: 2.0 m/s.
Track: Three joints along the drop.

Answer: _____
_____
_____

10. (6 points) Solve this case and explain your reasoning: Target 52 s. Practice run: 50.6 s. The calibration table says each screw turn adds 0.4 s. What should the team do?

Answer: _____
_____
_____

11. (6 points) Solve this case and explain your reasoning: Calibration: base path 34 s; each slope-screw turn adds 0.8 s; removable spiral adds 9 s. Plan for a Target Time of 46 s.

Answer: _____
_____
_____

12. (2 points) Define “Loop height” in the context of Roller Coaster.

Answer: _____
_____
_____

13. (2 points) Define “Run Score” in the context of Roller Coaster.

Answer: _____
_____
_____

14. (2 points) Define “Footprint” in the context of Roller Coaster.

Answer: _____
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_____

15. (2 points) Define “IMA < 1” in the context of Roller Coaster.

Answer: _____
_____
_____

16. (2 points) Define “Rolling resistance” in the context of Roller Coaster.

Answer: _____
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17. (2 points) Define “Calibration table” in the context of Roller Coaster.

Answer: _____
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18. (2 points) Define “Slope adjuster” in the context of Roller Coaster.

Answer: _____
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